

**Reverse-Engineering Inductive Bias:
A Hierarchical Bayesian Account of Overhypotheses**

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The corresponding presentation for this article can be accessed at <https://jlschnatz.github.io/hbm/>. Open materials, including the code to reproduce the results of the paper and presentation are available at <https://github.com/jlschnatz/hbm>.

Abstract

Humans possess a remarkable ability to form rich generalizations from sparse data. This efficiency is driven by inductive biases that restrict the hypothesis space during the learning process. This article describes how hierarchical Bayesian models provide a computational, reverse-engineering account of this cognitive capacity. By operating across multiple levels of abstraction, these models formalize how learners exploit the hierarchical structure of their environment to form overhypotheses, which are higher-level expectations that systematically guide subsequent, lower-level inferences. To build intuition, a hierarchical Beta-Binomial model is first used to demonstrate how population-level regularities, such as homogeneous marble colours within bags, enable precise predictions from sparse observations in a simplified marble colour-guessing game. This framework is then extended to a multidimensional Dirichlet-Multinomial model to formalize the acquisition of the shape bias in infant category learning. By closely aligning with empirical data from toddlers in a noun generalization study, the model illustrates how overhypotheses can be acquired through experience. In this way, hierarchical Bayesian models offer a nuanced perspective between nativist and empiricist views of cognition, showing how learners can generalize efficiently from experience when constrained by plausibly innate priors. Importantly, while these models primarily operate at the computational level of analysis, bridging the gap to algorithmic and implementation levels remains important for a comprehensive understanding of human cognition.

Keywords: Hierarchical Bayesian Model, Inductive Bias, Overhypothesis, Category Learning

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Reverse-Engineering Inductive Bias: A Hierarchical Bayesian Account of Overhypotheses

Humans are remarkably efficient learners. Even when exposed to very little data, we can form rich expectations about our environment and make accurate predictions. For example, upon hearing a novel word applied to a single unfamiliar object, a toddler effortlessly learns to extend that new label to other objects of the same kind. This capability is even more remarkable when considering that there are nearly infinitely many possibilities regarding which aspects of the object define the new entity, such as its color, shape, material, or even its spatial location. In the most extreme case, this may even manifest as *one-shot learning*, the ability to make precise inferences about future data after observing only a single instance of a new concept.

To navigate this vast space of possibilities, human cognition fundamentally relies on the notion of *abstraction* (Griffiths et al., 2024; Kemp, 2008). Knowledge is not represented as a flat collection of isolated facts, but is instead organized across different levels of a hierarchy. This concept of hierarchical structure is ubiquitous throughout human cognition. One classic example is the syntactical organization of language, where continuous, low-level speech signals are organized into phonemes, which combine to form words, and are ultimately structured into phrases and sentences governed by syntactic grammatical rules (Griffiths et al., 2024).

Within this hierarchical structure, learning operates through simultaneous inferences across multiple levels, continuously integrating bottom-up data processing with top-down expectations (Kemp, 2008). What makes this system so efficient is the *inductive bias* it affords. For any meaningful generalization to occur from sparse data, learners must rely on this abstract background knowledge to restrict the hypothesis space they consider (Griffiths et al., 2024). In concrete terms, this constraining mechanism is realized through an *overhypothesis* – a piece of abstract knowledge or assumptions that define and structure the available hypothesis space at a lower, less abstract level (Kemp et al., 2007; 2006).

While the theoretical necessity of overhypotheses is clear, understanding their underlying mechanism requires a formal framework. For a reverse-engineering account of cognition, the key challenge is to explain the nature, function, and origin of this abstract knowledge in precise computational terms (Griffiths et al., 2024). This approach seeks to understand the mind by first identifying the core computational problems it faces, such as the problem of induction, and then demonstrating how observed human behavior naturally arises from solving those problems mathematically. Drawing on the theoretical framework by Griffiths et al. (2024), this article describes how hierarchical Bayesian models formalize the concepts of abstraction, inductive bias, and overhypotheses. These models provide a reverse-engineering account of human cognition by demonstrating how abstract prior knowledge is acquired through experience, and how the resulting overhypotheses efficiently guide future learning. To build a mathematical intuition, the article first lays out the simplest possible example, a hierar-

chical *Beta-Binomial model* illustrating how overhypotheses about color distribution of bags of marbles are formed during a marble guessing game. Next, this framework is generalized to a real-world empirical application, demonstrating how a multidimensional *Dirichlet-Multinomial model* explains the acquisition of the shape bias in early human category learning.

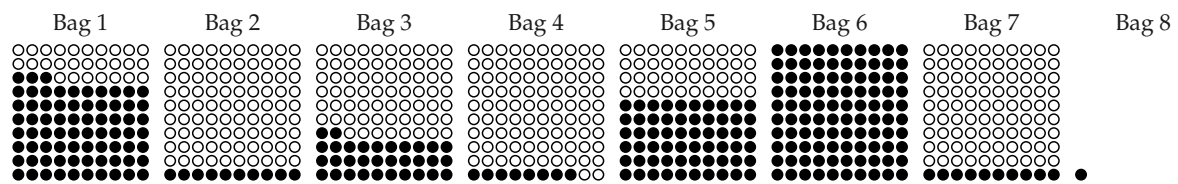
1 The Hierarchical Beta-Binomial Model

1.1 Building an Intuition for Inductive Bias

Imagine a fictitious scenario in which two children play a color guessing game with marbles from different bags. One child draws marbles sequentially from each bag, while the other tries to predict their color (either black or white). As shown in Figure 1.1, 100 marbles are drawn from each of seven different bags, revealing their respective color distributions. The task of the child is now to predict the color of the second marble drawn from the eighth bag after the first one has been revealed, given the observed distributions of the previous seven bags.

Figure 1.1

Color Distribution of Bags of Marbles in the Example



Note. Each bag in this example contains 100 marbles that are either black or white. The proportion of black marbles varies between bags. The task is to predict the color of the next marble in bag eight after observing the first marble and all the other seven bags. Figure generated using the *ggplot2* (Wickham et al., 2026) R-package.

At first glance, a single black marble contains very little information about the color of future marbles. And yet, the guessing child feels surprisingly confident. Based on the patterns observed in the previous bags and the first draw from the eighth bag, it intuitively seems far more likely than chance that the second marble will also be black. What gives rise to this remarkably accurate intuition?

Initially, a naive explanation might suggest that the child can make such accurate predictions simply because they were informed by prior data. However, merely observing the seven bags of marbles does not, on its own, allow for precise inferences. To exemplify, imagine that all marbles were poured into a single giant container and all but the last marble were drawn. Since the overall color distribution across bags is roughly equal, the container would be filled half with white and half with black marbles and thus the child's prediction would be no better than chance. The key difference between these two examples lies in the *hierarchical structure* of the data. As introduced earlier, human learning operates by making inferences across multiple levels of abstraction. Here, marbles are grouped within bags, and each bag has its own color distribution. By recognizing that each bag has a high probability of containing mostly one color,

the child is able to leverage this higher-level information to make more accurate predictions about individual marbles at the lower level.

While this conceptual explanation accounts for the child’s intuition, building a reverse-engineering account requires a mathematical formalization of the problem. The following section translates the marble game into a probabilistic model to show how the observed behavior naturally arises from solving the underlying problem of induction.

1.2 Formalization of the Marble Game

The formalization begins at the lowest level and then progresses up the hierarchy, ultimately constructing a probabilistic model that reflects both marble-level outcomes and bag-level structure (see Griffiths et al., 2024). The lowest-level of the model starts with the data itself (i.e., marbles drawn from one or more bags). For each bag i , y_i black marbles are independently drawn from n_i total marbles, which can be represented as the set d_i .

$$d_i : \{y_i, n_i\} \quad (1)$$

To formalize the data-generating process, probability distributions are used to describe how these observations arise. In the literal marble example, marbles are drawn without replacement, which would imply a changing probability for a black across draws, which would formally correspond to a Hypergeometric distribution. However, the hierarchical Bayesian model adopts a more abstract representation. Each bag is assumed to be characterized by an underlying probability of “producing” a black marble, and draws are treated as conditionally independent given this parameter. This corresponds either to sampling with replacement or to an infinitely large bag. Under this representation, the number of black marbles observed in n_i draws follows a *Binomial* distribution.

$$y_i | n_i \sim \text{Binom}(\theta_i) \quad (2)$$

The parameter θ_i represents the probability of drawing a black marble from bag i . Importantly, the subscript i implies that this probability may vary across bags. At this level in the hierarchy, a Bayesian model would estimate and quantify uncertainty of each θ_i separately. Consequently, information about the color distribution in one bag would not inform inference for another bag. In the example of the two children, predictions for the eighth bag would therefore rely exclusively on the single observed marble from that bag. Such a model would ignore the hierarchical structure present in the data and would lead to highly uncertain predictions.

To account for this structure, it is assumed that the individual parameters θ_i are themselves drawn from a broader population distribution (Griffiths et al., 2024). This higher-level distribution represents the general characteristics of bags in terms of their color composition and allows information to be shared across bags. Because this distribution is defined over a

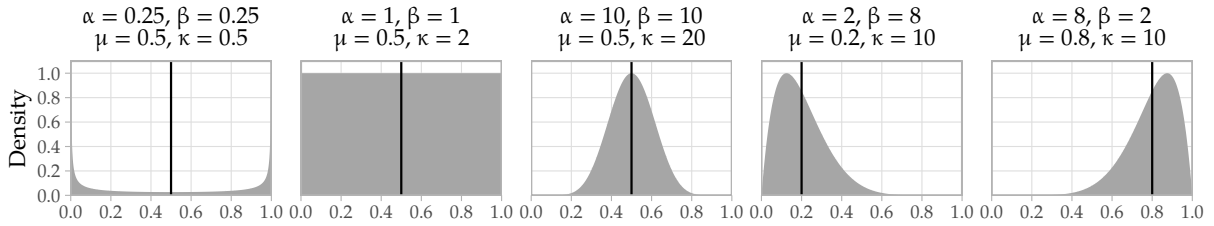
probability parameter $\theta_i \in [0, 1]$, a natural choice is the *Beta* distribution. The Beta family is defined on the unit interval and is sufficiently flexible to represent a wide range of beliefs about how likely bags are to contain black marbles (see Figure 1.2). Formally, this second level of the hierarchy can then be written as

$$\theta_i \sim \text{Beta}(\alpha, \beta) \quad (3)$$

The Beta distribution in its standard parameterization is defined by the two real-positive valued ($\mathbb{R}^+$) shape parameters α and β . These parameters can be interpreted as prior pseudo-counts. More specifically, before observing the current data, the model behaves as if it had previously observed $\alpha - 1$ black marbles and $\beta - 1$ white marbles. In this way, they encode both prior expectations and the strength of prior belief. When $\alpha = \beta = 1$ the Beta distribution reduces to the Uniform distribution, corresponding to complete uncertainty regarding θ_i (see Figure 1.2). As α and β increase, the distribution becomes more concentrated around its mean, reflecting greater prior certainty. Depending on which parameter is greater relative to the other, the skewness of the distribution shifts corresponding to varying beliefs in θ_i . When $\alpha + \beta < 2$, the distribution is U-shaped, placing more mass near 0 and 1 (see Figure 1.2).

Figure 1.2

Shape of the Beta Distribution for Different Parameter Values



Note. α and β : shape parameters, $\mu = \frac{\alpha}{\alpha+\beta}$: mean parameter, $\kappa = \alpha + \beta$: concentration parameter. The black line indicates the mean of each distribution. Figure generated using the *ggplot2* (Wickham et al., 2026) and *ggdist* (Kay & Wiernik, 2025) R-packages.

Because this parametrization is not completely intuitive in the marble game example, the distribution can be reparametrized in terms of a prior mean $\mu = \frac{\alpha}{\alpha+\beta}$ and a concentration parameter $\kappa = \alpha + \beta$. The former parameter represents the literal mean of the distribution (black line in Figure 1.2) and the concentration parameter how strongly the prior concentrates around the mean (i.e., the effective prior sample size). This means that the precision parameters is inversely related to the variance of the Beta distribution (exact relationship $\sigma^2 = \frac{\mu(1-\mu)}{1+\kappa}$).

Speaking in terms of the example, the mean parameter μ can be interpreted as the average probability of drawing a black marble across the entire population of bags. The precision parameter κ describes how homogeneous or heterogeneous the color distribution of the marbles is within each bag. When $\alpha + \beta < 2$, there is a high probability that the colors will be homogenous in color within bags, meaning they will be either almost exclusively white or

black (like in Figure 1.1). In contrast, when the precision parameter κ is large the distribution is tightly centered around its mean μ , and the bags are likely to contain a mixture of black and white marbles with proportions close to μ .

The final layer of the model quantifies uncertainty of the bag-level parameters μ and κ by placing hyperprior distributions on each parameter. In this example, a Uniform distribution is placed over the mean μ and an Exponential distribution over the precision κ (Griffiths et al., 2024).

$$\begin{aligned}\mu &\sim \text{Uniform}(0, 1) \\ \kappa &\sim \text{Exponential}(1)\end{aligned}\tag{4}$$

A uniform prior on μ reflects complete prior uncertainty about the average probability of drawing a black marble across bags. Consequently, every admissible values of μ (between 0 and 1) is conceived as equally likely. In comparison to the prior on μ , the exponential prior with rate $\lambda = 1$ on κ is *informative*, because it biases the expectation towards smaller precision values. Therefore, it places more prior certainty on scenarios with low variability within bags and high variability between bags, including near-homogeneous bags when κ is very small. In summary, before seeing any data, the children consider all expected proportions of black marbles across all bags equally plausible, while anticipating substantial variability in the bag-specific probabilities of drawing a black marble.

Importantly, the rate parameter λ of the hyperprior distribution is assumed to be known in advance. In principle, however, additional layers can be added to hierarchical models by once again placing prior distributions over these parameters. This may be especially relevant for learning processes that require more complex hierarchies, for example, language comprehension (Griffiths et al., 2024; Kemp, 2008). To ensure the model is identifiable, it is necessary to assume that the hyperparameters in the highest levels of the hierarchy are a priori known.

Posterior inference of the parameters at any level of the hierarchical model, conditional on the observed data, can be obtained by combining the prior distributions with the data likelihood using Bayes' rule (Gelman, 2014):

$$P(\theta, \alpha, \beta | \mathbf{y}) \propto \underbrace{P(\alpha, \beta)}_{\text{Hyperprior}} \cdot \underbrace{P(\theta | \alpha, \beta)}_{\text{Conditional prior}} \cdot \underbrace{P(\mathbf{y} | \theta, \alpha, \beta)}_{\text{Likelihood}}\tag{5}$$

Inferences for any bag-level parameter θ_i can be obtained by integrating out the hyperparameters α and β . Because the resulting integral is typically analytically intractable, numerical integration via Markov Chain Monte Carlo (MCMC) algorithms is used as an approximation (Griffiths et al., 2024).

$$P(\theta_i | d_1, \dots, d_n) = \iint P(\theta_i | \alpha, \beta, d_i) P(\alpha, \beta | d_1, \dots, d_n) d\alpha d\beta\tag{6}$$

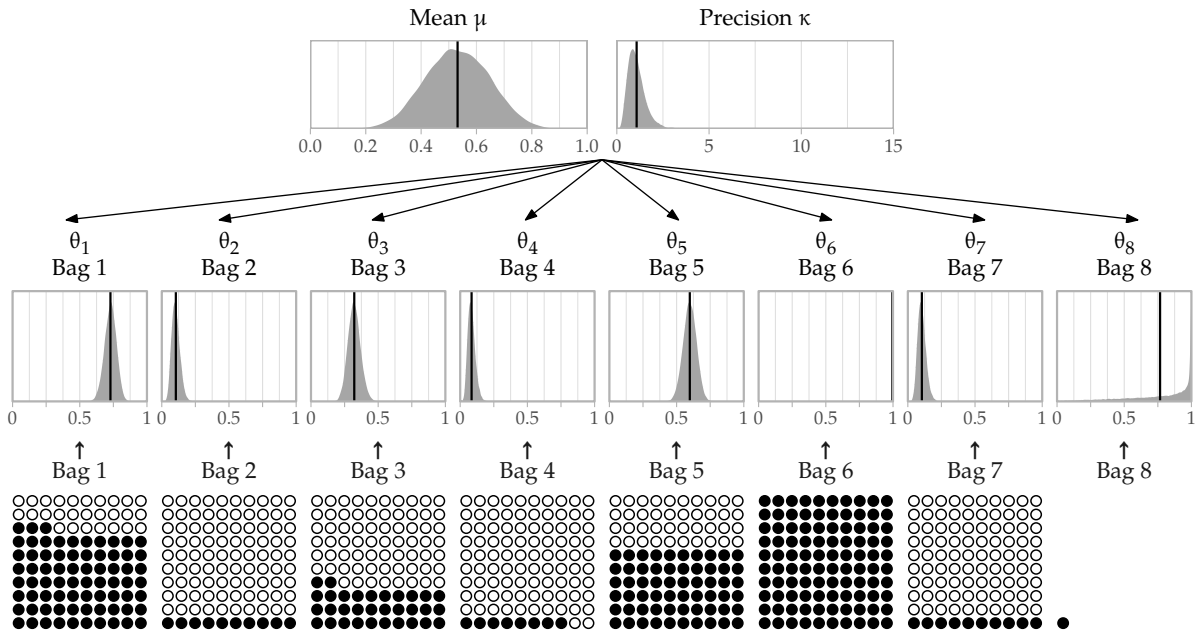
1.3 Model Application and Posterior Inference

Figure 1.3 visualizes the estimated posterior distributions for the bag-level parameters θ_i as well as the shared population-level parameters μ and κ , after observing the data shown in Figure 1.1 and applying the model described in Equation 2 – Equation 4. Most importantly, the posterior distribution for θ_8 demonstrates that the model’s predictions align with the intuition of the children. Although only a single black marble has been observed in bag 8, the model is highly confident that θ_8 is close to one.

How does the model achieve such precise predictions? The key lies in the highest level of the hierarchy. The posterior distribution of κ places high certainty on low values of the parameter, inducing a U-shaped population-level distribution over θ_i . This implies that the bags tend to be internally homogeneous in color. By capturing this population-level regularity, the model mathematically formalizes the acquisition of an overhypothesis. This learned overhypothesis acts as a highly informative prior. It severely restricts the hypothesis space, allowing the model to leverage a single low-level observation to make a precise, confident prediction for a new bag.

Figure 1.3

Posterior Estimates Showing How Bag-Level Homogeneity Enables Accurate Predictions for a New Bag



Note. Gray ribbon represents the kernel density estimate from the posterior samples (methodological detail see section Section 4). Black solid lines indicate the posterior median. θ_i : probability of drawing a black marble. Inspired from Griffiths et al. (2024). Figure generated using the *gdist* R package (Kay & Wiernik, 2025).

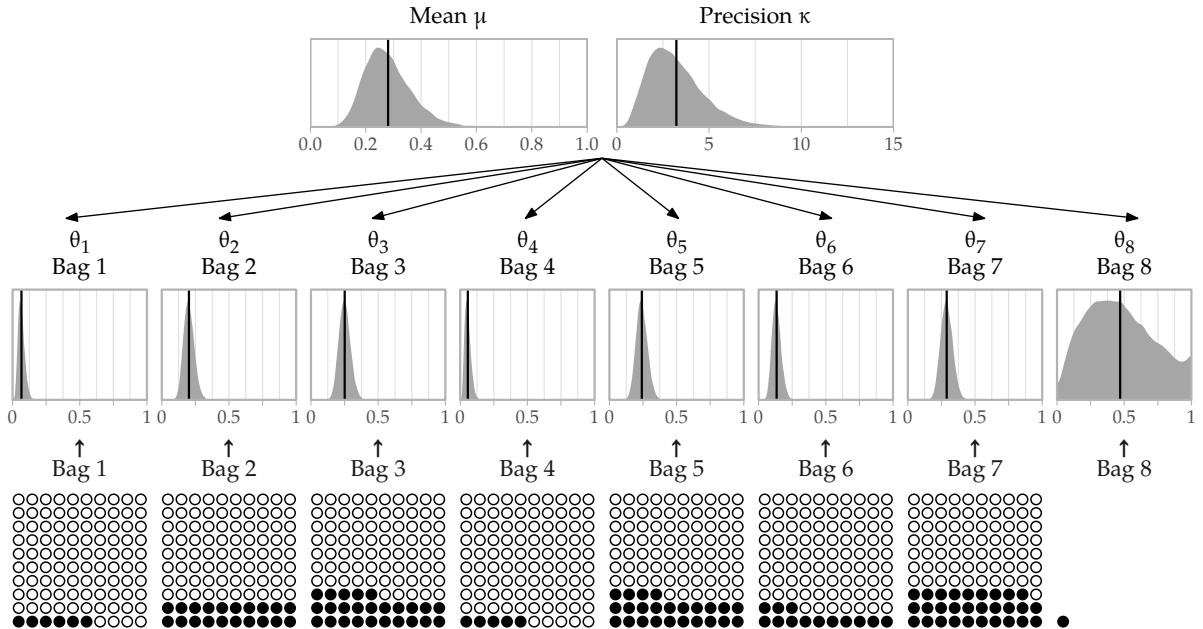
To contrast these results with a scenario in which precise predictions are not possible, Figure 1.4 shows the posterior distributions obtained from data that do not support strong inferences about the probability of drawing a black marble in bag 8. The posterior distribution

for θ_8 (second-to-bottom panel) illustrates this very clearly. The median value, indicated by the solid black line, suggests that the model’s prediction is no better than random chance, and the high uncertainty is reflected in the heavy tails of the distribution.

Examining the data together with the population-level distribution of bags reveals the reasons for this uncertainty. Overall, the bags contain a lower proportion of black marbles, and within each bag, the colors are more evenly mixed. At the level of the model parameters, two important changes are evident. First, the population-level mean μ is smaller, reflecting the lower average proportion of black marbles across all bags. Second, the precision parameter κ is higher than in the previous example, indicating that the model has learned that the bags are more heterogeneous in their color composition. Because the model has inferred that bags are heterogeneous, a single observation in bag 8 provides very little information about the bag-specific probability θ_8 . As a result, the posterior for θ_8 remains diffuse, and the model cannot make a precise prediction. Because the data lacks strong population-level regularities, the model cannot form an overhypothesis to meaningfully reduce the hypothesis space regarding the probability of a black marble at the lower level. This example illustrates an important property of hierarchical models that also applies to human cognition: the formation of an overhypothesis can only occur when strong population-level regularities exist at higher levels of abstraction. In their absence, predictions for new, sparsely observed bags remain uncertain.

Figure 1.4

Posterior Estimates Showing How Heterogeneous Bags Lead to Uncertain Predictions for a New Bag



Note. Gray ribbon represents the kernel density estimate from the posterior samples (methodological detail see section Section 4. Black solid lines indicate the posterior median. θ_i : probability of drawing a black marble. Inspired from Griffiths et al. (2024). Figure generated using the *ggplot2* (Wickham et al., 2026) and *ggdist* (Kay & Wiernik, 2025) R-packages.

2 Multidimensional Extension for Category Learning

The marble game example has conceptually illustrated how hierarchical structure in data is exploited by humans to form multiple layers of abstraction and how Bayesian models capture this process in a probabilistic framework. Additionally, the example has illustrated how Bayesian models formalize the idea of an overhypothesis when higher-level regularities exist, which in turn can support precise predictions at lower levels even in one-shot learning scenarios. While the marble game is a simplified example, the same principles apply to more complex, real-world learning tasks. In particular, hierarchical Bayesian models provide a natural framework for understanding the acquisition of shape bias that guides category learning in children.

2.1 The Shape Bias Phenomenon

To exemplify, consider a scenario involving a dyadic interaction between a mother and her child. When the mother points to a novel object on a counter that she refers to as a *pen*, the child faces an immediate inductive challenge: Upon which characteristics should the concept of *pen* be generalized to identify future instances? While the child could theoretically extend the label to other objects sharing the same material, color, texture, or spatial location, empirical evidence suggests that children predominantly extend the label to objects that share the same shape (eg., Smith et al., 2002). This phenomenon is known as the *shape bias* (Griffiths et al., 2024), which can be conceptualized as a learned overhypothesis defined by a higher-level abstract expectation that members of a category tend to share a similar shape. By forming this overhypothesis, the child drastically reduces the hypothesis space upon which perceptual characteristics an unknown category may be generalized.

2.2 Formalizing the Shape Bias

To formally model and explain the shape bias phenomenon from a Bayesian perspective, the hierarchical beta–binomial model introduced in the section Section 1.2 must be extended in several respects. Table 2.1 provides an overview of the modified components. At the lowest level of the hierarchy, the observed data consist of counts of object exemplars (e.g., a pen lying on the counter). The grouping variable that defines the hierarchical structure is the object category to which a given exemplar belongs (e.g., pen). In contrast to the marble example, however, object exemplars are not characterized by a single binary feature dimension (e.g., black vs. white). Instead, they vary along multiple feature dimensions, such as shape, color, texture, and size. Moreover, each of these dimensions typically comprises more than two and arguably infinitely many possible levels (e.g., a small red plastic pen vs. a large aluminum pen).

Table 2.1*Model Adaption of the Marble Game Example to the Shape Bias in Category Learning*

Example	Hierarchy	Data	Dimensionality	Type	Values
Marble game	Bags	Marbles	Unidimensional	Color	Binary
Shape bias	Object category	Exemplars of objects	Multidimensional	Shape, color, texture, size	Multinomial

More formally, let there be m feature dimensions (shape, color texture, size, etc.) and i object categories. Within each feature dimension m there are F_m possible levels (Glassen & Nitsch, 2016). For instance, in the category *pen* ($i = 1$) along the color dimension ($m = 2$), if there are $F_2 = 3$ possible colors, a child may observe that out of 10 exemplars, 5 were black, 3 white and 2 silver. Rather than a binomial distribution, these counts follow its generalization, the *Multinomial* distribution.

$$\mathbf{y}_{im} \sim \text{Multinomial}(\boldsymbol{\theta}_{im}) \quad (7)$$

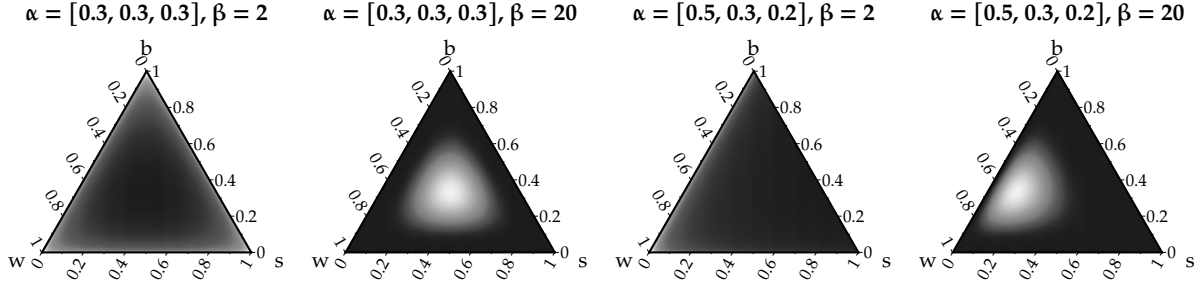
The parameter vector $\boldsymbol{\theta}_{im}$ is a *simplex* (i.e., a vector summing to 1) representing the proportions of feature levels. In the example case, the empirical proportions of colors would be $\theta_{11} = [0.5, 0.3, 0.2]$ for each color, respectively. The conjugate prior distribution of the Multinomial is a *Dirichlet* distribution with parameters α_m and β_m , the multivariate generalization of the Beta distribution from the marble game example.

$$\boldsymbol{\theta}_{im} \sim \text{Dirichlet}(\alpha_m \boldsymbol{\beta}_m) \quad (8)$$

Intuitively, α_m captures the extent to which exemplars within a category are uniform in their feature values, while β_m represents the expected distribution of feature values across the entire category (Kemp et al., 2007; 2006). In other words, β_m encodes the population-level expectation, and α_m determines how strongly individual exemplars conform to that expectation. Linking this back to the marble example, α_m and β_m can be thought of as a generalization of the mean μ and precision κ , respectively. The ternary plot in Figure 2.5 shows the Dirichlet distribution for three possible feature values. In the left column, the three colors (black, white, silver) are equally likely across categories (see α_m), but the small value of β (analogous to κ) implies that object exemplars within categories are mostly uniform in color. In the second column, the overall color expectation remains the same, but a larger β increases within-category variability, making exemplars more heterogeneous. The two rightmost columns illustrate the same concept for non-uniform overall color expectations (i.e., different category probabilities for colors).

Figure 2.5

Visualization of the Dirichlet Distribution for Three Feature Values of the Color Dimension



Note. Shading indicates the probability density over the simplex; lighter regions correspond to more likely combinations of the three color probabilities. w: white, b: black, s: silver (see main text). Figure generated using the *ggplot2* (Wickham et al., 2026) and *ggttern* (Hamilton & Ferry, 2018) R-packages.

As in the marble game example, prior distributions are required for α_m and β_m . Specifically, the concentration parameter α_m is assigned an exponential prior with a fixed rate of $\lambda_m = 1$ and the population-level feature proportions β_m follow a Dirichlet prior with uniform weights (i.e., multivariate uniform distribution).

$$\begin{aligned}\alpha_m &\sim \text{Exponential}(1) \\ \beta_m &\sim \text{Dirichlet}(\mathbf{1}_{F_m})\end{aligned}\tag{9}$$

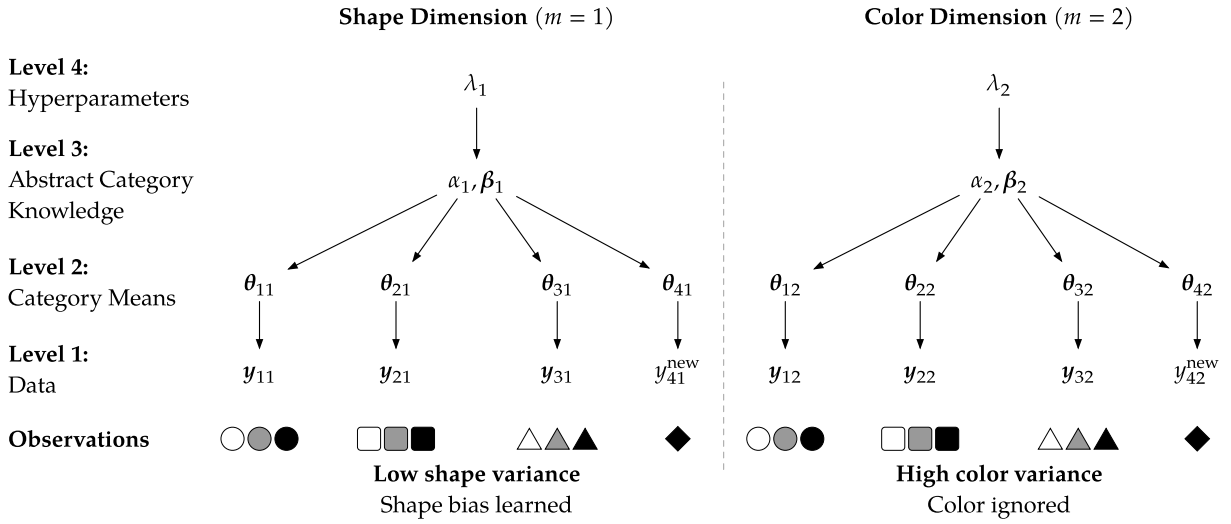
The multivariate uniform prior reflects complete prior uncertainty about feature-value proportions for each dimension, while the exponential prior with rate $\lambda = 1$ favors relatively homogenous features distributions within categories. This completes the extension of the marble game framework to multiple feature dimensions, yielding the *Dirichlet–Multinomial model* (Griffiths et al., 2024).

How does this formalization help to explain the shape bias in human category learning? To answer this, the model described in Equation 7 – Equation 9 is instantiated independently for each feature dimension m . Figure 2.6 illustrates a simplified learning scenario involving two feature dimensions: shape ($m = 1$) and color ($m = 2$). Here, the learner observes three established object categories, each containing three exemplars, alongside a single exemplar (y^{new}) belonging to a novel category. Each dimension possesses three possible feature values: circle, square, or triangle for shape, and white, gray, or black for color. Crucially, the training data reveal a structural regularity: exemplars within the same object category consistently share the same shape (i.e., the shape bias), but they vary in color. For the color dimension (right panel), the presence of multiple colors within each category indicates high within-category heterogeneity. Consequently, the model infers a relatively large value for α_2 (conceptually analogous to the broader distribution in the second column of Figure 2.5). In contrast, for the shape dimension (left panel), objects within a category are perfectly homogeneous. This causes the

model to infer a small value for α_1 , reflecting tight within-category precision (analogous to the first column of Figure 2.5). By inferring a small α_1 and a large α_2 , the model effectively forms the higher-level overhypothesis that object categories are strictly organized by shape, but not by color. When the learner encounters the exemplar y^{new} of an unknown object category, which happens to be a black diamond, this learned overhypothesis serves as an informative prior expectation. Because the hypothesis space has been drastically reduced along the shape dimension, the model confidently infers that future instances of this new category will also be diamond-shaped, but it remains appropriately agnostic about their color. In doing so, the hierarchical framework mathematically replicates the shape bias.

Figure 2.6

Hierarchical Structure in the Dirichlet–Multinomial Model Explaining the Shape Bias



Note. Simplified representation of the Dirichlet–Multinomial model across two feature dimensions. Three established categories (three exemplars each) and one novel category (single exemplar, y^{new}) are shown. The model infers a small concentration parameter for the homogeneous shape dimension (α_1) and a large parameter for the heterogeneous color dimension (α_2), establishing the overhypothesis that drives the shape bias for novel objects.

2.3 Empirical Application in Noun Generalization

The Dirichlet–Multinomial model formally defines the computational problem of category learning, conceptualizing the shape bias as an overhypothesis acquired through experience. However, a reverse-engineering account also requires demonstrating that human behavior naturally approximates this ideal solution. To test this, Griffiths et al. (2024) applied the model to a noun generalization task (see also Kemp et al., 2007; 2006), drawing on data from and inspired by a landmark developmental study by Smith et al. (2002).

The original study by Smith et al. (2002) investigated how toddlers learn to generalize novel object names. When young children are presented with a novel object and a novel name, they face an immediate inductive challenge: deciding which perceptual properties are relevant for extending the category. To test whether children learn to systematically attend to shape

through experience, the researchers conducted a 9-week longitudinal study with 17-month-old toddlers. At this age, children are typically too young to systematically extend object names based on shape.

To evaluate whether a hierarchical Bayesian model can capture this learning process, the training environment of the experiment by Smith et al. (2002) can be formally represented as a data matrix (see Table 2.2). The model is exposed to training data consisting of two exemplars (e_1 and e_2) from each of four distinct object categories. The objects vary along four feature dimensions: shape, texture, color, and size. For this implementation, it is assumed that the number of possible feature values F_m is 10 for shape, texture, and color, and 2 for the size dimension. Importantly, as shown in the first row of Table 2.2, the model’s training data reflects the actual experiment in that object categories are perfectly invariant in their shape, but heterogeneous across the other dimensions.

Table 2.2

Training and Testing Data Based on a Study by Smith et al. (2002)

Feature m	Category 1		Category 2		Category 3		Category 4		Target	Choices		
	e_1	e_2	e_1	e_2	e_1	e_2	e_1	e_2	x_{dax}	t_1	t_2	t_3
m_1 : Shape	1	1	2	2	3	3	4	4	5	5	6	6
m_2 : Texture	1	2	3	4	5	6	7	8	9	10	9	10
m_3 : Color	1	2	3	4	5	6	7	8	9	10	10	9
m_4 : Size	1	2	1	2	1	2	1	2	1	1	1	1

Note. Each category has two exemplars e_1 and e_2 . The target object must be matched (x_{dax}) to three previously not seen test objects $t_1 - t_3$ with unknown true object category, see also Griffiths et al. (2024).

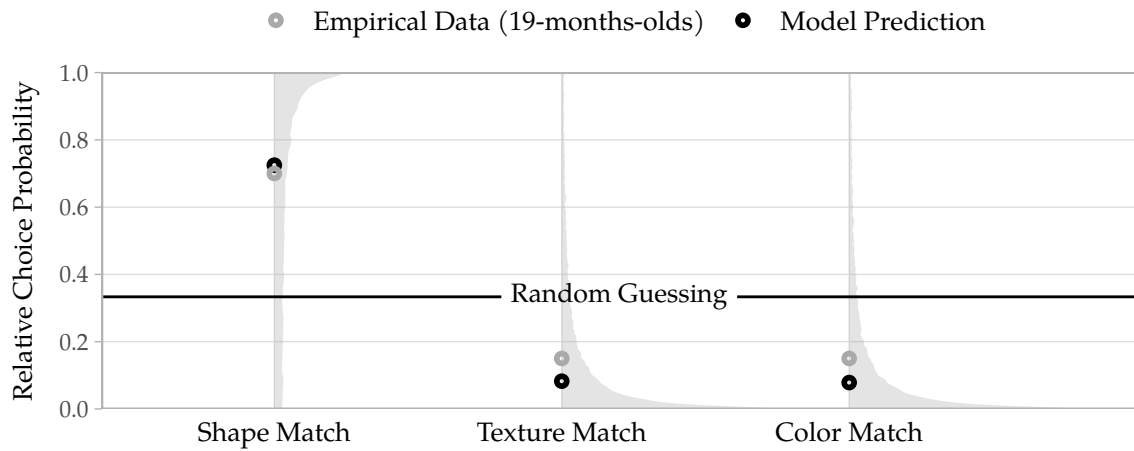
Following the training phase, the model is subjected to a second-order generalization test designed to mirror the empirical task given to the toddlers. The model is presented with a novel target object, arbitrarily labeled a “dax” (x_{dax}), belonging to a completely new object category ($i = 5$) that was not present during training. The task is to determine which of three novel test objects (t_1, t_2 , or t_3) is most likely to also be a “dax”. The test objects are carefully chosen, such that t_1 matches the target x_{dax} only in shape, t_2 matches only in texture, and t_3 matches only in color. This experimental design forces the model to rely on its higher-order generalizations to resolve the competing feature matches.

Figure 2.7 illustrates the posterior distribution of the model’s relative choice probabilities for matching each of the three test objects to the novel target, x_{dax} . Given the three available choices, the baseline probability for random guessing is $p = \frac{1}{3}$. The posterior distribution reveals a high degree of certainty that the model will select t_1 , the test object that matches the target in shape. This strong preference, alongside the fact that the probabilities for the texture and color matches fall well below chance, clearly demonstrates the model’s successful acqui-

sition of an overhypothesis (i.e., the shape bias). Furthermore, the figure plots the empirical data from Smith et al. (2002), showing that the model’s posterior belief almost perfectly mirrors the aggregate choice frequencies of the trained toddlers. This alignment is particularly compelling because a baseline group of children in the same study, who received no prior category training, failed to generalize by shape and instead chose among the test objects at random (Smith et al., 2002). By successfully replicating these developmental trajectories, the hierarchical Bayesian framework provides a reverse-engineering account of category learning and demonstrates how shape bias emerges through experience as a learned overhypothesis.

Figure 2.7

Comparison of Model Predictions and Empirical Data for a Novel Noun Generalization Task



Note. Lightgray ribbon: kernel density estimate of the posterior samples, black point: softmax-normalized posterior median of model. gray point: empirical choice percentage of 19-month-old children based on data from Smith et al. (2002). Random guessing probability of $p = \frac{1}{3}$. Figure generated using the *ggplot2* (Wickham et al., 2026) R-package.

3 Synthesis and Discussion

Humans possess a remarkable ability to form rich generalizations from sparse data, enabling highly efficient learning. Drawing on the work by Griffiths et al. (2024), this article has illustrated how hierarchical Bayesian models provide a reverse-engineering account of this cognitive capacity. Specifically, these models probabilistically capture how human learners exploit the hierarchical structure of their environment to form overhypotheses. By acting as higher-level prior expectations, these overhypotheses significantly reduce the hypothesis space at lower levels of abstraction. Ultimately, this structural restriction not only enables rapid learning for novel instances, but also provides a formal explanation for both how humans “learn to learn” and where prior distributions actually originate (Griffiths et al., 2024).

These principles were first illustrated in simplified marble color guessing game. Formalized through the beta-binomial model, this example demonstrated how population-level regularities in the form of nearly homogenous marbles within bags enabled the formation of

a structural overhypothesis. In turn, this higher-order expectation acted as a strong prior, enabling precise predictions about the color of future marbles within a novel, sparsely observed bag. Crucially, the Bayesian model also captured the formal boundary condition of this process. When such higher-level regularities do not exist (e.g., if previously observed bags are completely heterogeneous), no informative overhypothesis can be established. Consequently, the model’s prior remains diffuse, and inference at the lower level remains uncertain.

Building on this simple example, the framework was extended to a Multinomial-Dirichlet model to demonstrate its real-world relevance in infant category learning. Specifically, this application formalized the acquisition of the shape bias as a learned overhypothesis. Just as the simpler model detected color uniformity within bags, this extension demonstrated how learners extract structural regularities across multiple feature dimensions, namely, the realization that objects within the same object category are organized by shape rather than texture or color. Furthermore, the model’s inferences aligned closely with empirical data from a noun generalization study inspired by Smith et al. (2002). Because the trained toddlers in this study were otherwise too young to systematically exhibit a shape bias, this alignment reinforces the idea that such overhypotheses are not innate, but are instead learned through experience.

This insight also bears on the longstanding debate between nativist and empiricist accounts of cognitive development. Hierarchical Bayesian models offer a nuanced perspective within this debate. On the one hand, this article has shown that inductive biases, such as the shape bias, can emerge through experience as learners form overhypotheses about the structure of their environment. On the other hand, the formalization of both examples has illustrated that probabilistic inference cannot occur in a vacuum, and that these models must bake in fixed prior assumptions at the highest level of abstraction. Ideally, the hierarchy in the models can be iteratively extended until the remaining prior knowledge corresponds only to assumptions that are plausibly innate (Griffiths et al., 2024).

Beyond these specific examples, hierarchical Bayesian models possess broad applicability across many cognitive domains. Their relevance is evident whenever a learning problem can be decomposed into multiple layers of abstraction, as seen in, for example, in causal learning, language acquisition, vision, and property induction (Griffiths et al., 2024).

Despite these explanatory strengths, it is important to recognize the specific level of abstraction at which these models operate. Following Marr (2010) tripartite framework, the hierarchical Bayesian approach represents “function-first” reverse-engineering strategy that characterizes cognition primarily at the computational level (Griffiths et al., 2024). This level defines the abstract problem the system solves and identifies its mathematically optimal solution, rather than specifying the specific procedures or data structures (algorithmic level), or describing neural instantiation (implementation level). Crucially, as Griffiths et al. (2024) emphasize, offering an explanation in terms of Bayesian inference does not imply that individuals are explicitly calculating Bayes’ rule in their heads. Instead, the assumption of optimality

serves as a methodological framework to understand behavior as a rational response to environmental structures. While this article highlighted the utility of a purely computational approach, bridging the gap between this level of analysis and its algorithmic and neural implementation remains a critical challenge for future research to achieve a comprehensive understanding of human cognition.

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4 Appendix

4.1 Implementation of the Beta-Binomial Model

Simulation of the two examples in section Section 1.3 was carried out using the R programming language (R Core Team, 2025), version 4.5.2. Bayesian hierarchical model estimation was conducted in *Stan* (Carpenter et al., 2017) via Hamiltonian Monte Carlo (HMC) using the No-U-Turn Sampler (NUTS), implemented through the *cmdstanr* (Gabry et al., 2025) interface. Four Markov chains were run with 5000 warm-up iterations and 10000 post-warm-up samples per chain. The control parameters were set to a target average acceptance probability of `adapt_delta = 0.99` and a maximum tree depth of `max_treedepth = 15`. The population parameters were $\varphi = \{0.8, 20\}$, $\mu = \{0.5, 0.2\}$ for the dispersion and location parameter of the Beta distribution, respectively. Marble data were simulated according to Equation 2 - Equation 4. The *Stan* code is provided below.

```
/* Beta-Binomial Hierarchical Model in STAN (Vectorized) */
data {
  int<lower=0> N;          // Number of bags
  array[N] int<lower=0> n; // Number of marbles drawn from each bag
  array[N] int<lower=0> y; // Number of black marbles in each bag
}

parameters {
  real<lower=0, upper=1> mu; // Mean of Beta
  real<lower=0> phi;         // Precision of Beta
}

transformed parameters {
  // Precompute alpha and beta for Beta-Binomial
  real alpha = mu * phi;
  real beta = (1 - mu) * phi;
}

model {
  // Hyperpriors
  mu ~ beta(1, 1); // uniform prior
  phi ~ exponential(1);

  // Likelihood (Beta-Binomial)
  y ~ beta_binomial(n, alpha, beta);
}

generated quantities {
  // posterior samples for theta
  vector[N] theta;
  for (i in 1:N) {
    theta[i] = beta_rng(alpha + y[i], beta + n[i] - y[i]);
  }
}
```

4.2 Implementation of the Dirichlet-Multinomial Model

The Dirichlet-Multinomial model was fitted using the same software and settings as the Beta-Binomial model, with training data detailed in Table 2.2. For the model implementation, it was assumed that each of the M total feature dimensions possesses $F = 10$ possible values¹. The model places an Exponential prior on the concentration parameter α_m and a uniform Dirichlet prior on the base rate simplex β_m .

$$\alpha_m \sim \text{Exp}(1.0) \quad \beta_m \sim \text{Dir}(\mathbf{1}_F) \quad (10)$$

To ensure stable HMC sampling, the latent category distributions $\theta_{j,m}$ were marginalized out, and the observed counts $\mathbf{y}_{j,m}$ of $n_{j,m}$ total objects were evaluated via a marginalized Dirichlet-Multinomial likelihood.

$$\mathbf{y}_{j,m} \mid \alpha_m, \beta_m \sim \text{DirMultinomial}(n_{j,m}, \alpha_m \beta_m) \quad (11)$$

Because $\theta_{j,m}$ were marginalized out during inference, their posterior samples were analytically reconstructed post-hoc using the Dirichlet-Multinomial conjugacy in the generated quantities block.

$$\theta_{j,m} \mid \mathbf{y}_{j,m}, \alpha_m, \beta_m \sim \text{Dir}(\alpha_m \beta_m + \mathbf{y}_{j,m}) \quad (12)$$

To simulate forced-choice noun generalization, the model computes the posterior predictive probability that test object k matches the novel target “dax”. For each dimension m , this probability is formulated as a ratio. The numerator captures feature-specific evidence. A binary indicator I for an exact match ($f_{k,m} = f_{\text{dax},m}$) is combined with the prior pseudo-counts $\alpha_m \beta_{m,f_{k,m}}$. This is normalized by the denominator, which sums the total evidence of the single observed “dax” (1) plus the total prior weight (α_m).

$$P(f_{k,m} \mid f_{\text{dax},m}, \alpha_m, \beta_m) = \frac{I_{f_{k,m}=f_{\text{dax},m}} + \alpha_m \beta_{m,f_{k,m}}}{1 + \alpha_m} \quad (13)$$

To mirror the mutually exclusive behavioral task, absolute log-probabilities across all dimensions are summed and softmax-normalized into relative choice probabilities p_k .

$$p_k = \text{softmax} \left(\sum_{m=1}^M \ln P(f_{k,m} \mid f_{\text{dax},m}, \alpha_m, \beta_m) \right) \quad (14)$$

¹This is a slight departure from the theoretical parameterization described by Griffiths et al. (2024), who posited 10 possible values for shape, color, and texture, but only 2 for size. Because the probabilistic programming language *Stan* does not natively support ragged arrays (i.e., varying bounds for F_m), strictly enforcing this constraint would require a complex reparameterization using a Gamma-Dirichlet equivalence trick. Given that assuming a uniform dimensionality of $F = 10$ across all features does not alter the model’s core qualitative predictions regarding the shape bias, this simplification was adopted for computational clarity.

```

/* Dirichlet-Multinomial Hierarchical Model in Stan */
data {
  // Training data
  int<lower=1> N; // Categories (4)
  int<lower=1> M; // Feature dimensions (4)
  int<lower=1> F; // Number of possible values (10)
  // Train data
  array[N, M, F] int<lower=0> y;
  // Test data for the generalization task
  array[M] int<lower=1, upper=F> x_dax;
  array[3, M] int<lower=1, upper=F> x_choices;
}

parameters {
  array[M] real<lower=0> alpha;
  array[M] simplex[F] beta;
}

model {
  alpha ~ exponential(1.0);
  for (m in 1:M) {
    beta[m] ~ dirichlet(rep_vector(1.0, F));
    for (j in 1:N) {
      // marginalized likelihood (integrate out theta)
      y[j, m, ] ~ dirichlet_multinomial(alpha[m] * beta[m]);
    }
  }
}

generated quantities {
  // Reconstruct category-specific distributions
  array[N, M] simplex[F] theta;
  for (m in 1:M) {
    for (j in 1:N) {
      theta[j, m] = dirichlet_rng(alpha[m] * beta[m] + to_vector(y[j, m, ]));
    }
  }
  // Compute noun generalization choice probabilities
  vector[3] log_prob_choice;
  vector[3] choice_prob;
  for (k in 1:3) {
    log_prob_choice[k] = 0;
    for (m in 1:M) {
      int f_choice = x_choices[k, m];
      real prior_count = alpha[m] * beta[m, f_choice];
      real observed_count = (f_choice == x_dax[m]) ? 1.0 : 0.0;
      real p = (observed_count + prior_count) / (1.0 + alpha[m]);
      log_prob_choice[k] += log(p);
    }
  }
  // Softmax normalization
  choice_prob = softmax(log_prob_choice);
}

```

Statutory Declaration

I herewith declare that I have composed the manuscript myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The manuscript in the same or similar form has not been submitted to any examination body and has not been published. This manuscript has not yet, even in part, been used in another examination or as a course performance. In addition, I acknowledge that I have used artificial intelligence tools during the preparation of this manuscript. I am aware that the use of machine-generated texts does not guarantee the quality of content and text. I therefore declare that I have only used text-generating AI tools as an aid and that my creative influence predominates in this work. All intellectual decisions, argumentation, and conclusions presented in this manuscript are my own or cited appropriately.

AI Tool	Purpose
DeepL	Language translation, grammar checking
Gemini Pro 3.1	Copyediting, text refinement, Stan translation of the noun generalization task.

Date, Place

Signature